

Visual Perception in VR Training: Impact of Information Transfer Methods

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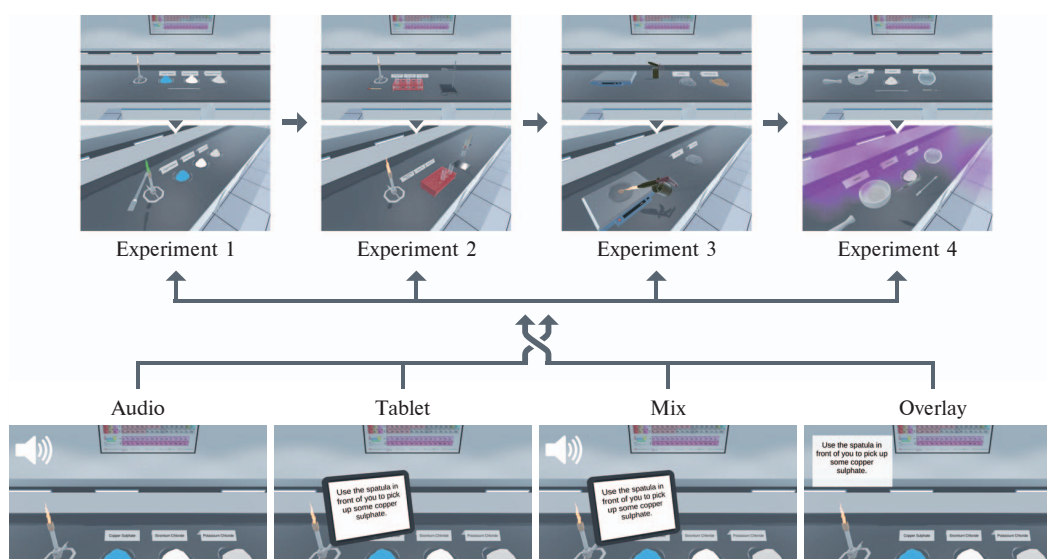


Figure 1: The upper row depicts the chemical experiments that the participants perform sequentially, while the lower row shows the information transfer methods that are randomly distributed across the chemical experiments.

ABSTRACT

Virtual Reality (VR) has great potential for education and training, but it is not fully understood how to make VR training as effective as possible. An investigation indicated that a combination of auditory and textual information is the best way to present training instructions and explanations to the user. However, is it also suitable to perceive the training content visualized in the virtual environment? We conduct a within-subjects study to investigate this by comparing users' gaze attention in four information transfer methods. The results are consistent with previous findings that the auditory-visual combination is the most appropriate of the methods compared.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality; Human-centered computing—Human computer interaction (HCI)—HCI theory, concepts and models; Applied computing—Education—Interactive learning;

1 INTRODUCTION

Virtual Reality (VR) offers a wide range of application areas, but one of the most promising is education and training. In the virtual environment, users are not exposed to physical boundaries or dangers, so it is possible to experience medical diseases and their treatment in a virtual patient [1]. However, it is also possible for civilians to learn ordnance disposal [3]. In these, users receive the relevant training information and interaction instructions differently. For single-user applications, an investigation has shown that combining text and

audio information is the most effective and preferred method for subjective ratings [2]. Here, the conditions *Audio* (speech only), *Tablet* (text on a virtual object mapped to the controller position), *Mix* (combination of audio and tablet), and *Overlay* (text overlaid on the user's field of view) were compared with each other. In this paper, we build on this work and investigate the effects of these methods on users' gaze attention. The gaze should reveal how the information transfer method affects the user's ability to pay attention to the information presented in the virtual environment.

2 METHODOLOGY

We compare the four information transfer methods using a within-subjects study. For this purpose, we design and implement four training scenarios, including all four information transfer methods. The training content of these scenarios deals with high school-level chemistry experiments that are comparable in their complexity, interaction, and effort. The participants' task involves executing the instructions, observing the resulting reactions, and comprehending the explained theoretical chemistry. In addition, the information transferred is identical for all information transfer methods, leading to redundant content in the *Mix* condition. We used the HTC Vive Pro Eye head-mounted display and Unity3D. Overall, this methodology reflects the previous work [2] on which we rely.

This work focuses on recognizing differences between the methods regarding gaze attention. Therefore, we classify the target of the participant's gaze as one of the following categories: 1) Environment Objects (e.g., wall, table, lamp, etc.), 2) Textual Instructions (displayed text on the tablet or overlay), 3) Interactive Objects (objects the participant interacts with, e.g., spatula, mortar, substances), and 4) Reactive Objects (the chemical reactions of the experiments, e.g., weight display, smoke).

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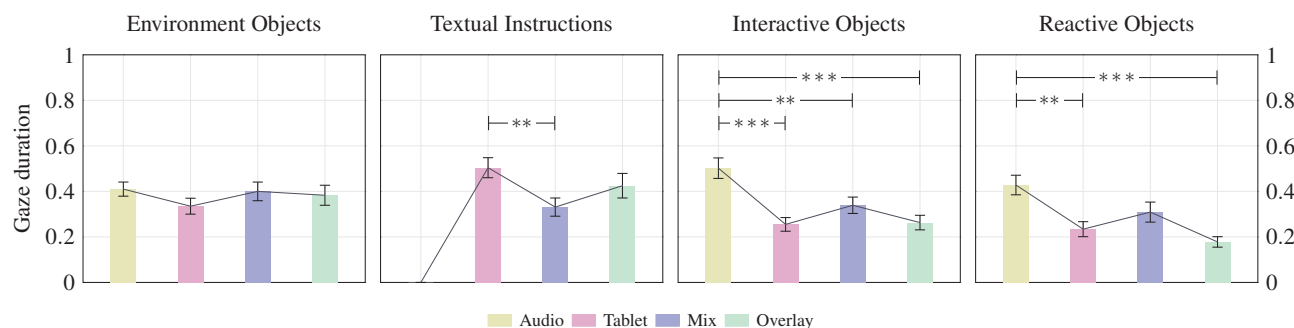


Figure 2: Gaze attention time on specific visual content, min-max normalized to ensure all training scenarios are comparable, scaled to [0..1]. Error bars represent the standard error.

For the study, we recruited 44 participants (34 male, 10 female) aged between 19 and 31 ($M = 24.39$, $SD = 2.92$). All participants were academics and completed the following procedure: 1) Consent form and demographic questionnaire, 2) Introduction to the hardware (with eye-tracking calibration) and software, 3) Tutorial scene, and 4) Main scenarios with the four information conditions. The order of these conditions has been counterbalanced.

3 RESULTS

We applied one-way repeated measures analysis of variance (ANOVA) and post-hoc paired t-tests with Bonferroni correction for statistical parametric data analysis. In cases where Mauchly's test indicated that the assumption of sphericity was violated, the degrees of freedom were corrected. For the comparability of the gaze attention, the durations (x) for each VR chemistry experiment were separately normalized as follows: $x_{norm} = \frac{x_i - x_{min}}{x_{max} - x_{min}}$. These min-max normalized results are presented in Figure 2 and were converted to a common scale with the range of [0, 1], where higher values indicate a longer duration.

Environment Objects The statistical analysis revealed no significant differences between the conditions $F(3, 129) = 1.217$, $p = .306$, $\eta_p^2 = .029$. Accordingly, the comparison between the conditions indicated that the differences between *Audio* ($M = .41$, $SD = .20$), *Tablet* ($M = .34$, $SD = .23$), *Mix* ($M = .40$, $SD = .27$), and *Overlay* ($M = .38$, $SD = .28$) corresponded approximately to the same amount of attention the subjects paid to the environmental objects.

Textual Instructions The *Audio* condition was excluded from the statistical analysis since no textual instructions existed. This analysis indicated a significant difference between the conditions $F(2, 86) = 4.667$, $p = .012$, $\eta_p^2 = .102$. A Bonferroni-adjusted pairwise comparison showed significance for the *Tablet-Mix* ($p = .009$) comparison. Overall, participants focused most on the instructions on the *Tablet* ($M = .50$, $SD = .29$), followed by the *Overlay* ($M = .43$, $SD = .35$) and the *Mix* ($M = .33$, $SD = .26$) conditions. No further significant effects were observed.

Interactive Objects We found significant differences in participants' attention to the virtual objects they interacted with $F(3, 129) = 11.704$, $p < .001$, $\eta_p^2 = .222$. The post-hoc test indicating that *Audio-Tablet* ($p < .001$), *Audio-Mix* ($p = .005$), and *Audio-Overlay* ($p < .001$) comparisons significantly differed. Interactive objects received the most attention in the *Audio* ($M = .50$, $SD = .29$) condition, followed by *Mix* ($M = .34$, $SD = .23$), *Overlay* ($M = .26$, $SD = .20$), and *Tablet* ($M = .26$, $SD = .20$).

Reactive Objects Attention to the chemical reactions resulting from specific events also revealed significant differences between the four conditions $F(3, 129) = 8.412$, $p < .001$, $\eta_p^2 = .170$. Pairwise comparisons revealed a significant difference between the *Audio-Overlay* ($p < .001$), and *Audio-Tablet* ($p = .002$) conditions. Here, followed by *Mix* ($M = .31$, $SD = .29$), *Tablet* ($M = .23$, $SD = .21$), and *Overlay* ($M = .18$, $SD = .15$), participants paid the most attention to the reactions in the *Audio* ($M = .43$, $SD = .28$) condition.

4 DISCUSSION AND CONCLUSION

Findings show that participants spent approximately the same time looking at Environmental Objects for all methods.

With the Textual Instructions, users of the *Tablet* condition invest significantly more time looking at the text than with the *Mix* method. This indicates that participants follow the *Audio* instructions in the *Mix* modality. The results also show that users of the *Audio* condition have problems appropriately processing all the audio information since the *Mix* method participants also focused on the text instructions. This is in line with previous results [2], which showed that users of the *Audio* condition were less able to follow the information than those using the *Tablet* and *Mix* methods.

As users of the *Audio* method have no readable information, they can invest more time viewing the virtual environment. However, the Interactive and Environmental Objects results show they use the free visual capacities for focusing on context-related objects.

The Reactive Objects are the most relevant in all four chemistry experiments, as they visualize the chemical reactions. Participants with the *Tablet* and *Overlay* method pay the least attention to Reactive Objects.

Overall, the *Mix* and *Audio* conditions offer users of VR training the best opportunity to perceive the visual training information. Considering that users prefer the *Mix* method and see it as the most effective method, the *Mix* method should be used for comparable VR training scenarios because effective VR training is only valuable when it corresponds to the user's expectations [2].

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